

SUBASH ADHIKARI

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August 5, 2025

Dear Dr. Jinke Tang,

I am writing to apply for the position of Assistant Professor in the Department of Physics and Astronomy at the University of Wyoming's College of Engineering and Physical Sciences. I learned about this exciting opportunity through the AAS job register and am enthusiastic about the possibility of joining your department.

I received my Ph.D. in Computational Space Plasma Physics from the University of Delaware in August 2022. My graduate and ongoing research uses massively parallel Particle-In-Cell (PIC) simulations, primarily using NCAR/UCAR supercomputers, to explore space plasma phenomenon such as magnetic reconnection and plasma turbulence. Additionally, I analyze observational data from some of the spacecraft such as Parker Solar Probe (PSP) and the magnetospheric multiscale mission (MMS). This combination of numerical modelling of space plasmas along with observational data analysis overlaps with some of the Astrophysical plasma phenomenon and positions me uniquely for the advertised role.

My proposed research centers on understanding plasma turbulence as a function of heliospheric distance, role of shear flows in magnetic reconnection and plasma turbulence, and energy transfer characteristics in electron only reconnection. These plasma processes have significant implication for particle heating and energization in the Earth's magnetosphere and the solar wind, where such processes are commonly observed.

I am positive my research expertise can establish a new and complimentary subfield within your department contributing to its research profile and broader mission. I am also eager to bring my experience and passion into the classroom, mentoring and motivating students and fostering a collaborative research environment.

Thank you for your time and consideration. Feel free to contact me at (302) 310-7755 or subash@udel.edu if you require any additional information or have any questions regarding my application. I look forward to hearing from you soon.

Best regards,

Subash Adhikari